The change of variable from Schrodinger to QFT Document

The value of R is a constant distance

The onduscular equation $\frac{2m}{\hbar^2}(E-U(r))rf(r,t)+\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (1) is valid in both spherical and cylindrical coordinates; it emerges from a combination of Schrodinger, which is the first degree with respect to time, and the Klein-Gordon temporal equation of the second degree with respect to time. Also, we chose a parameter R, thus for a small radius $r_{Bohr,}$ for the atomic value to coincide with the Schrodinger equation exactly and predict the evolution of the wave function to huge values of r about the galactic distances. The wave function of the Schrodinger equation has the supposition that is null at infinity, is normed, has the dimension of $[L^{-3/2}]$, and predicts the probability of the existence of the microparticle in volume dV. Schrodinger's equation is (2):

$\frac{2m}{\hbar^2}(E-U(r))f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=ih\frac{\partial f(r,t)}{\partial t}$ & Klein-Gordon is $\frac{2m}{\hbar^2}f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (3)

Thus, Eq. (1) for m = 0 is a wave equation with variable parameters and seems to be true. $\frac{R}{r}\frac{\partial^2(rf(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 rf(r,t)}{\partial t^2}$ (4) But all 3 equations are different, and the wave equation is not satisfactory

$$\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (5),$$

1.1 Step 1 — Energy balance for a bound electron

For a hydrogenic electron in a Coulomb field, the non-relativistic energy is $E=\frac{1}{2}m_eV^2-\frac{e^2}{4\pi\varepsilon_0 r}+\frac{l(l+1)\hbar^2}{2m_er^2}$. (1.1)

Solving for the kinetic term and dividing by $\hbar^2/(2m_e)$ gives the "wavenumber-squared" combination

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_er^2}\right]=\frac{m_e^2V^2}{\hbar^2}=k^2(r). \quad (1.2)$$

1.2 Step 2 — Onduscular wave ansatz

In TauTimp, an electron is represented by a spatial-temporal wave $f(r,t)$ propagating at the local "ondoscopic" speed V, with the cosmological scale R entering through a curvature-corrected Laplacian. The local d'Alembertian satisfies

$$\Box f=\frac{R}{r^2}\frac{\partial^2(rf)}{\partial r^2}-\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}=0. \quad (1.3)$$

The prefactor R/r^2 encodes the cosmological coupling: at $r=R$ the operator reduces to the standard ∂_r^2 on the conformal coordinate, while at atomic scale $r\ll R$ it amplifies the radial derivatives by $R/r^2\sim 10^{22+22}=10^{44}$, which is exactly the factor needed to make $E/\hbar$ and $k^2(r)$ commensurable in the combined equation below.

1.3 Step 3 — Coupling the wave equation to the energy condition

Combining (1.2) and (1.3) and adding the local Coulomb + centrifugal source on the LHS:

$$\boxed{\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_er^2}\right]f(r,t)+\frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.} \quad (1)$$

This is the **onduscular (TauTimp) electron equation**, in its raw form before separation of variables. With the cosmological radius $R=3.9228\times10^{22}$ m and the boundary condition $g(R)=0$ (where $f(r,t)=g(r)\,u(t)$), it determines the bound spectrum.

1.4 Step 4 — Separation of variables

Let $f(r,t)=g(r)\,u(t)$ with $u(t)=\cos(\omega t)$, so that $\partial_t^2u/u=-\omega^2$. The energy condition $E=\hbar\omega$ relates the temporal frequency to the electron energy. Substituting into (1):

$$\frac{2m_e}{\hbar^2}\left[E+\frac{e^2}{4\pi\varepsilon_0 r}-\frac{l(l+1)\hbar^2}{2m_er^2}\right]g(r)+\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2}=-\frac{\omega^2}{V^2}\,g(r).$$

The RHS contributes a term $-\omega^2/V^2 \cdot g$ that combines with the energy term on the LHS. For non-relativistic bound states, $V^2 = 2E/m_e$, so $\omega^2/V^2 = E^2 m_e/(2\hbar^2 E) = E m_e/(2\hbar^2)$, which is already absorbed into the $2m_e E/\hbar^2$ term. The radial equation becomes $\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0\,r} - \frac{l(l+1)}{r^2}\right] = 0,$ (2)

where we used $2m_e e^2/(\hbar^2 \cdot 4\pi\varepsilon_0) = 2/a_0$.

This is the **original radial ODE TauTimp** used for the Taylor comparison.

The complete onduscular equation, in the form that reproduces the entire atomic structure of hydrogen, is

$$\boxed{\begin{aligned}&\frac{2m_e}{\hbar^2}\left[E + \frac{e^2}{4\pi\varepsilon_0\,r} - \frac{l(l+1)\hbar^2}{2m_e r^2} + \xi(r)\,\langle \vec{L}\cdot\vec{S}\rangle + V_{HF}(r)\,\langle \vec{S}_e\cdot\vec{I}_p\rangle\right] f(r,t)\\ &\qquad + \frac{R}{r^2}\frac{\partial^2(r\,f(r,t))}{\partial r^2} \;=\; \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}.\end{aligned}} \qquad (3)$$

with - $\xi(r) = \frac{1}{2m_e^2 c^2\,r}\frac{dV}{dr} = \frac{\hbar\alpha}{2m_e^2 c}\frac{1}{r^3}$ for Coulomb $V(r) = -\hbar c\alpha/r$, - $V_{HF}(r) = \frac{2\mu_0}{3}\,g_p\,\mu_B\,\mu_N\,\delta^3(\vec{r}) \cdot \frac{4}{\hbar^2}$ — the Fermi electron–proton contact term.

Equation (3) is the centerpiece of this document.

2. The two governing equations (for comparison with Schrödinger) 2.1 Schrödinger radial equation (hydrogen, reduced form $u = r\psi$)

$$\mathcal{L}_S\,u(r) \;\equiv\; \frac{d^2u}{dr^2} + u(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] \;=\; 0. \qquad (4)$$

2.2 ODE TauTimp (original operator, scalar, no spin)

From (2):

$$\frac{R}{r^2}\frac{d^2(r\,g(r))}{dr^2} + g(r)\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0\,r} - \frac{l(l+1)}{r^2}\right] = 0. \qquad (5)$$

With $u(r) = r\,g(r)$, the kinetic term becomes $(R/r^2)\,u''(r)$. Multiplying by r^2/R to clear singularities:

$$u''(r) + \frac{u(r)}{R}\cdot r\left[\frac{2m_e E}{\hbar^2} + \frac{2}{a_0 r} - \frac{l(l+1)}{r^2}\right] = 0.$$

This is the form used in the Taylor comparison below.

3. Taylor comparison at $r = 0$

Multiply (4) by r^2 and (5) (in the $u = rg$ form) by r^2/R to clear $1/r$ and $1/r^2$. Substitute $u(r) = \sum_{n\geq 0} u_n\,r^n$ with the regularity boundary condition $u_0 = 0$ (so that $g(r) = u(r)/r$ is finite at the origin).

3.1 Coefficient table (at $l = 0$, regular solution $u_0 = 0$)

N	$[r^N]$ of Schrödinger $\times\, r^2$	$[r^N]$ of TauTimp $\times\, r^2/R$
0	0 (since $u_0 = 0$)	0
1	$2u_0/a_0 = 0$	$2u_0/(a_0 R)\cdot R = 0$
2	$2u_2 + 2u_1/a_0 + \frac{2m_e E}{\hbar^2}u_0$	$\frac{1}{R}[2u_2 + 2u_1/a_0]$
3	$6u_3 + 2u_2/a_0 + \frac{2m_e E}{\hbar^2}u_1$	$\frac{1}{R}[6u_3 + 2u_2/a_0]$
⋮	⋮	⋮
6	$30u_6 + 2u_5/a_0 + \frac{2m_e E}{\hbar^2}u_4$	$\frac{1}{R}\left[30u_6 + 2u_5/a_0 + \frac{2m_e E}{\hbar^2}u_0\right]$

3.2 Conclusion of the Taylor comparison

- r^0 **coefficient**: equal (both zero) — guaranteed by the regularity condition $u_0 = 0$. ✓
- r^1, r^2: equal trivially because $u_0 = 0$.

- **Divergence at r^3**: Schrödinger has the energy term $\frac{2m_eE}{\hbar^2}u_1$; TauTimp does not (it appears only at r^6 and is additionally suppressed by $1/R$).

Net effect: the original TauTimp ODE misses the energy term $\frac{2m_eE}{\hbar^2}$ at atomic scale; the centrifugal term carries $1/R$ instead of $1/r^2$.

This is the equation for any micro particle that we take for granted. We obtain equation (1) also for v = c, thus m = 0, we have equation (5).
The effective wave function of the onduscular equation has the supposition as null at R (or 2R) is normed, has the dimension of [$L^{-3/2}$], and predicts the probability of the existence of the micro particle in volume dV. Also, a condition of the derivative = R in origin verifies the condition that the speed of interaction has the velocity V, whatever it will be. The value of R = $3.922833072\cdot10^{22}$ m, thus 4.146526391 millions of light-years, is calculated in wes.pdf
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Thus, we have:

$$R = \frac{1+\frac{L^2}{r_b\hbar^2}}{8\pi\varepsilon} = \frac{1+\frac{l(l+1)}{r_b}}{8\pi\varepsilon} = \frac{r_b + l(l+1)}{r_b 8\pi\varepsilon} \cong \frac{l(l+1)}{r_b 8\pi\varepsilon}$$

Where rb is the radius Bohr, and ε is the vacuum permittivity, and the value of "l" is the angular momentum quantum number, should be 21. In the periodic table with 118 elements that we know, the principal quantum number "n" varies from 1 to 5, but theoretically, n should vary from n=1… ∞ (infinity), and the link to "l" is that l varies from 0… n-1. Is a little peculiar that the value of "l" should be 21, thus the principal quantum number n = 22, but it fits the dowsing measurement and gives a proper value to R equal to R = $3.922833072\cdot10^{22}$ m, thus 4.146526391 million light-years.

I do not know what the significance of l=21 is that fits the calculus, and probably there is another equation that gives a more accurate value of R, but it is a good start that fills the gap in the theory of a limited Local Universe & the initial conditions for the onduscular theory.

Numerical check at low values (order of Bohr radius) "The Wave Equation Scale",
1. Basic equations (Maple notations: h = ħ)
The Onuscular Equation (eq6 from Maple):

$$\text{eq6} := \frac{2m}{\hbar^2 R}\left(E + \frac{ec^2}{\rho} - \frac{L^2}{2m\rho^2} - \frac{\hbar^2 k}{2m}\right)\rho^2 f(\rho) + 2\frac{d}{d\rho}f(\rho) + \rho\frac{d^2}{d\rho^2}f(\rho) = 0$$

$$\frac{2m}{\hbar^2}\left(E + \frac{ec^2}{\rho} - \frac{L^2}{2m\rho^2} - \frac{\hbar^2 k}{2m}\right)\frac{\rho^2 f(\rho)}{\boldsymbol{R}} + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

Schrödinger equation (eq1 in Maple, section two):

$$\frac{2m}{\hbar^2}\left(E - \frac{ec^2}{\rho}\right)\rho f(\rho) + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

2. Variable change and scaling (rb = Bohr radius)
Substitutions shall apply:

$$\rho = r_b \cdot r, \quad f(\rho) = g(r), \quad E = \frac{m\,ec^4}{32\pi^2\varepsilon^2\hbar^2}E_0, \quad r_b = \frac{4\pi\varepsilon\hbar^2}{m\,ec^2}$$

The scaled Schrödinger equation becomes (eq2, final form):

$$\left(E_0 - \frac{8\pi\varepsilon}{r}\right)r\,g(r) + \frac{d^2(r\,g(r))}{dr^2} = 0$$

Scaled onnduscular equation (onduscular eq2):

$$\left(\frac{E_0 r^2}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2 r^2}{m^2 ec^4 R} + \frac{8\pi\varepsilon\, r}{R} - \frac{L^2}{\hbar^2 R}\right)rb\cdot g(r) + 2\frac{dg}{dr} + r\frac{d^2 g}{dr^2} = 0$$

3. Changing variables with the limit factor (border condition)
The zero transformation to r = 2R is applied:

$$f(\rho) = g(r)\cdot\frac{R - r\cdot r_b}{2R}$$

After substitution and simplification (eq2 → eq3 in Maple), the equation becomes:

$$r\frac{d^2g}{dr^2}+2\frac{dg}{dr}+\mathcal{T}(r)\cdot g(r)=0$$

where the term free expands in Taylor series around :$\mathcal{T}(r)r=0$

$$\mathcal{T}(r)=\underbrace{\frac{r_b(8\pi R\varepsilon-2)}{2R}}_{\text{ord. }0}+\underbrace{\frac{r_b(E_0R\,r_b-8\pi\varepsilon\,r_b)+\frac{r_b^2(8\pi R\varepsilon-1)}{R}}{2R}\cdot r}_{\text{ord. }1}+\underbrace{\frac{-r_b^3E_0+\frac{r_b^3(2E_0R^2-1)}{R^2}}{2R}\cdot r^2}_{\text{ord. }2}$$

4. The condition of overlap with Schrödinger

Comparing the coefficients with the scaled onuscular equation:

$$\frac{E_0r^2}{R}-\frac{16\hbar^4k\pi^2\varepsilon^2r^2}{m^2ec^4R}+\frac{8\pi\varepsilon r}{R}-\frac{L^2}{\hbar^2R}$$

3 equations are obtained by identifying the coefficients at each order in: r

Order	Matching equation
r^0 (constant)	$\frac{r_b(8\pi R\varepsilon-2)}{R}=-\frac{L^2}{\hbar^2R}$
r^1 (linear)	$\frac{r_b(2E_0R\,r_b-8\pi\varepsilon r_b)+\frac{r_b^2(8\pi R\varepsilon-1)}{R}}{R}=-\frac{8\pi\varepsilon}{R}$
r^2 (quadratic)	$\frac{1}{2}\cdot\frac{-r_b^3E_0+\frac{r_b^3(2E_0R^2-1)}{R^2}}{R}=\frac{E_0}{R}-\frac{16\hbar^4k\pi^2\varepsilon^2}{m^2ec^4R}$

5. Analytical solution wes.mw (31)

Maple solves the system of 3 equations in and gets: R, k $\boxed{R=-\frac{2r_b\hbar^2-L^2}{8\pi\varepsilon\hbar^2r_b}}$

or equivalent (eq. 32 of Maple): $R=\frac{-r_bh^2+L^2}{8\pi\varepsilon h^2r_b}$

where h = ħ (Maple notation).

The simplified form (Maple equation 37), for the orbital quantum number:l $\boxed{R=\frac{l(l+1)}{8\pi\varepsilon\,r_b}}$

This results from the identity, as follows:$L^2=l(l+1)\hbar^2$

$$R=\frac{r_b\hbar^2+l(l+1)\hbar^2}{8\pi\varepsilon\hbar^2r_b}=\frac{l(l+1)+1\cdot(r_b/r_b)}{8\pi\varepsilon\,r_b}\approx\frac{l(l+1)}{8\pi\varepsilon\,r_b}\quad(l\gg1)$$

A more rigorous calculation of the value of the constant distance R in the supposition of identity of spectral lines of Hydrogen, in the Schrodinger model, and the onduscular equation, so that the error corresponds to a precision $\varepsilon\approx10^{-64}$ — 62 orders smaller than the finest spectroscopic measurement.

$$\boxed{R(n_1,n_2,\varepsilon)=\frac{\pi\,a_0\,n_1\,n_2}{\sqrt{\varepsilon}}}$$

$$R=\frac{\pi\,a_0\,n_1n_2}{\sqrt{\varepsilon}}$$

$$R_{\text{opt}}(\varepsilon_{\text{rms}})=\frac{\pi\,a_0\sqrt{\langle n_1^2n_2^2\rangle}}{\varepsilon_{\text{rms}}}$$

With : $\sqrt{\langle n_1^2n_2^2\rangle}=9.959$, giving:

$$R_{\text{opt}}(\varepsilon_{\text{rms}})=\frac{\pi\cdot a_0\cdot9.959}{\varepsilon_{\text{rms}}}=\frac{1.656\times10^{-9}\text{ m}}{\varepsilon_{\text{rms}}}$$

So, we have R ≈ $3.9228\cdot10^{22}$ m link R.pdf

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